

Xingyi (Daniel) Chen

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RESEARCH INTERESTS

Statistical genomics and biomedical data science; development and evaluation of statistical methodology for high-dimensional genomic and biomedical data.

EDUCATION

Johns Hopkins University

Baltimore, MD

B.S. in Applied Mathematics & Statistics, minors in Computational Medicine and Mathematics

Expected May 2027

- GPA: 3.99/4.00 | Dean's List, 6 semesters
- Relevant Coursework:

Mathematics & Statistics: Real Analysis I/II (with Measure Theory), Probability, Mathematical Statistics, Applied Statistics & Data Analysis, Monte Carlo Methods, Optimization, Computational Mathematics (*in progress*), Multivariable Calculus, Linear Algebra, Differential Equations, Causal Inference (*audit*)

Computational Biology & Genomics: Computational Genomics: Data Analysis, Computational Biology & Bioinformatics, Algorithms for Genomic Sequence Data (*in progress*), Genomic Data Visualization, Computational Medicine, Systems Pharmacology, Translating Healthcare AI (*in progress*)

RESEARCH EXPERIENCE

Research Assistant – Hicks Lab

January 2025 – Present

Johns Hopkins University, Department of Biostatistics | PI: Stephanie Hicks, PhD

- Developed *IsoVita*, a statistical framework for identifying time-dependent isoform trajectories in bulk RNA-seq through gene-level feature condensation and parallel testing of discrete, continuous, and localized temporal patterns; performed algorithm design, simulation-based evaluation, and cross-dataset validation.
- Applied the framework to human hippocampal aging to identify age-associated isoform patterns; presented preliminary findings at Biology of Genomes, Cold Spring Harbor Laboratory, 2026.
- Developed and released *SpotSweeper-py*, a Python/scverse implementation of neighborhood-based quality control for spatial omics; validated concordance with the R/Bioconductor implementation, benchmarked computational performance and parameter sensitivity, and led the first-author F1000Research publication.
- Built reusable Visium HD spatial transcriptomics analysis infrastructure, including 10+ reproducible pipelines and the pip-installable *visiumhd-utils* package for preprocessing, QC, destriping, visualization, and figure generation, supporting an NIH R01 grant proposal.

Quantitative Sciences Undergraduate Research Experience (QSURE) Intern – Shah/McPherson Lab

New York, NY

Memorial Sloan Kettering Cancer Center | PI: Sohrab Shah, PhD; Mentors: Andrew McPherson, Matt Myers

Summer 2026

- Developed a deletion-specific one-sided scan statistic for low-coverage scDNA-seq, extending a multi-sample scan framework to test copy number 0 against the best-fitting nonzero state and aggregate evidence across cells.
- Benchmarked CBS, PELT, FPOP, dynamic programming segmentation, and scan-statistic approaches across sequencing depth, bin size, event length, aneuploidy, and subclonal prevalence, identifying failure modes and resolution limits.
- Calibrated the statistic using empirical null distributions and robustness analyses, characterizing recovery of rare subclonal deletions and sensitivity to bin alignment, carrier prevalence, and noise estimation.
- Built an end-to-end Python workflow for preprocessing, interval scanning, posterior carrier scoring, parallel computation, and visualization; applied it to MSK-SPECTRUM ovarian cancer scDNA-seq data to investigate candidate deletions and technical artifacts.

Independent Research Project – Donor-Aware scRNA-seq Evaluation

March 2026 – Present

Johns Hopkins University, Department of Biomedical Engineering | PI: Alexis Battle, PhD

- Led a donor-aware, multi-cohort benchmark of scRNA-seq cell-type annotation to quantify when and by how much random cell-level evaluation overestimates cross-donor generalization across atlas-scale and independent validation datasets.

- Compared random cell-level, donor-held-out, and cross-site transfer evaluation across HVG, PCA, Harmony, and scVI representations using macro F1, class-specific error analyses, and generalized linear mixed models to separate evaluation-scheme effects from donor- and batch-level heterogeneity.
- Characterized how task separability and cohort composition modulate the magnitude of performance inflation, including settings in which ceiling or floor performance can mask differences between random and donor-aware evaluation.
- Built a modular, reproducible Python benchmarking framework for multi-cohort analysis and cross-dataset validation; preparing a first-author preprint with reproducible code and outputs.

Research Assistant – Beer Lab

November 2023 – January 2025

Johns Hopkins University, Department of Biomedical Engineering | PI: Michael Beer, PhD

- Automated extraction and preprocessing of gene expression datasets from public databases and performed exploratory analyses and visualizations in R using tidyverse and ggplot2.

Bioinformatics Intern – Wei Lab

Shanghai, China

Lin Gang Laboratory | PI: Wu Wei, PhD

Summer 2024

- Reviewed and presented methodological summaries of five spatial-omics studies, highlighting statistics/ML approaches and potential follow-up analyses; supported two ongoing bioinformatics projects with data cleaning and exploratory analyses.

PUBLICATIONS & MANUSCRIPTS

[SpotSweeper-py](#) – spatially-aware quality control metrics for spatial omics data in the Python ecosystem.

Chen X, Totty M, Hicks SC. *F1000Research*. 2026; 15:33. DOI: [10.12688/f1000research.175162.1](https://doi.org/10.12688/f1000research.175162.1)

Chen X, Hicks SC. *IsoVita: A statistical method for detecting time-dependent isoform trajectories*. Manuscript in preparation.

Chen X, Li J, Battle A. *Evaluation design shapes estimates of cross-donor generalization in single-cell cell-type classification*. Manuscript in preparation.

PRESENTATIONS

Chen X, Myers M, McPherson A. *Identifying homozygous deletions in single-cell DNA sequencing data*. Oral presentation at *QSURE Symposium*, Department of Epidemiology & Biostatistics, Memorial Sloan Kettering Cancer Center, August 2026.

Chen X, Erdogdu B, Perteu M, Hicks SC. *Machine learning reveals tissue-agnostic and region-specific isoform aging markers in the human hippocampus*. Poster presentation at *Biology of Genomes*, Cold Spring Harbor Laboratory, May 2026.

TEACHING & LEADERSHIP EXPERIENCE

Undergraduate Lead Teaching Assistant

August 2024 – Present

Johns Hopkins University, Department of Mathematics | Supported Course: Differential Equations & Applications

- Promoted from Course Assistant to Lead TA; led weekly discussion section for 25 students and delivered 13 independently prepared instructional sessions per semester.
- Mentored undergraduate TAs, coordinated course logistics, co-developed quizzes/review sessions, and earned an average student teaching-evaluation score of 98%.

Co-Instructor & Peer Leader

August 2026 – Present

Johns Hopkins University | Supported Course: Intro to Hopkins: Arrive & Thrive

- Co-instruct a first-year seminar introducing incoming students to academic, professional, and student-support resources at Johns Hopkins.
- Mentor first-year students through their transition to university and facilitate discussions on academic success, community, and campus engagement.

SKILLS

Technical: Python, R, Bash; Git/GitHub, Unix/HPC/SLURM; Scanpy/AnnData, scikit-learn, Bioconductor; LaTeX, Jupyter/Quarto.